MIL-F-60542(MU)
14 June 1966

MILITARY SPECIFICATION

FUZE, MTSQ, M564E1 LOADING AND PACKING

1. SCOPE

1.1 This specification covers the loading and packing of the Fuze, MTSQ, M564E1 Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-314	- Waterproofness Test for Use in Development of Fuzes
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1235	- Single and Multilevel Continuous Samplings Procedures and Tables for Inspection by Attributes

DRAWINGS

F10542753	- Fuze, MTSQ, M564E1 Assembly
C8861213	- Box, Wirebound Packing, Ammunition for Artillery and Rocket Fuzes (within M2A1 Box)
F8864492	- Box, Ammunition, Metal, for Artillery Type and Rocket Fuzes
C8886458	- Box, Wirebound, Packing, Ammunition for Artillery and Rocket Fuzes (within M2A1 Box)

FSC 1390

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2.1(Cont'd)

DRAWINGS

C9214704	- Box, Wirebound, for Artillery Type or Rocket Fuzes
C9214705	- Carton, Packing, Ammunition for Artillery Type and Rocket Fuzes
IEL-10542753	- Inspection Equipment List for Fuze, M564E1 Assembly

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced standards and specifications.

3.2 First article sample.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.3 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.3.1 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

3.4 Waterproofness.- The fuze shall comply with MIL-STD-314.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample (not applicable).

4.3 Inspection provisions.

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4.3.1 Submission of product.- The product shall be presented in accordance with MIL-STD-105.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with MIL-STD-105. For classification of defects contained herein, the criteria in Table I shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing examination for visual defects.

TABLE I

Defect Classification	Acceptable Quality Level (AQL)
Major (Fuze)	AQL - 0.25 percent defective
Major (Packing and Marking)	AQL - 0.40 percent defective
Minor (Packing and Marking)	AQL - 0.65 percent defective

4.3.2.1 Fuze subassembly, Drawing F10542754, covering a detail of Drawing F10542753.

Defect	Method of Inspection
CRITICAL	
1. Safety adapter rotor armed or partially armed as viewed through the flash hole (check after removing dust cover)	Visual
MAJOR	
101. Insecure, improper or incomplete assembly	Manual and Visual
102. Moisture in flash hole	Visual
103. Foreign matter in flash hole	Visual
104. Fuze damaged or defective	Visual
MINOR	
None defined	

4.3.2.2 Closing cup assembly, Drawing B8595506, covering a detail of Drawing F10542753

CRITICAL

None defined

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Overall length exceeds maximum	Gage
102. Cup damaged or defective	Visual
MINOR	
None defined	

4.3.2.3 Fuze, MTSQ M564E1 assembly, Drawing F10542753.

CRITICAL	
1. Fuze not set of safe (S)	Visual
MAJOR	
101. Length from flange to bottom of cup exceeds maximum	Gage
102. Insecure, improper or incomplete assembly	Manual and Visual
103. Booster cup contacts shoulder of safety adapter	Visual
104. Cup damaged or defective	Visual
MINOR	
None defined.	

4.3.2.4 Box, ammunition, metal, Drawing F8864492.

CRITICAL	
None defined.	
MAJOR	
101. Moisture inside box (check prior to packing)	Visual
102. Gasket damaged or unseated (check prior to packing)	Visual
103. Polystyrene supports/fuzes packed inadequately	Visual and Manual
104. Dessicant bags/containers damaged, loose or missing	Visual and Manual
105. Box improperly closed and secured	Visual and Manual
MINOR	
201. Box damaged or deformed	Visual
202. Marking missing, incorrect or illegible	Visual

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4.3.2.5 Carton prior to sealing, Drawing C9214705.

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	Supports damaged or broken to extent items are loose or touch each other	Visual
MINOR		
201.	Packing component missing	Visual

4.3.2.6 Carton sealed, drawing C9214705.

CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Sealing strip torn, badly wrinkled or incomplete	Visual
202.	Carton torn, cut or punctured	Visual
203.	Contents move upon shaking carton	Visual
204.	Marking missing, misleading or unidentifiable	Visual

4.3.2.7 Box, wirebound, packing, ammunition, for Artillery Type and Rocket Fuzes (within M2A1 Box), Drawing C8861213.

CRITICAL		
	None defined	
MAJOR		
101.	Loop fasteners improperly assembled or broken	Visual

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4.3.2.7 (Cont'd)

Defect		Method of Inspection
MINOR		
201.	Marking, missing, incorrect or illegible	Visual
202.	Contents loose	Manual

4.3.2.8 Box, wirebound, packing, ammunition for artillery type and rocket fuzes (within M2A1 Box), Drawing C8886458.

CRITICAL

None defined

MAJOR

None defined

MINOR

201.	Contents loose	Visual
202.	Boxes improperly arranged	Visual
203.	Sleeve or spacer missing	Visual
204.	Strapping missing or loose	Visual
205.	Marking missing, incorrect or illegible	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Binder-lubricant content of tetryl.- A minimum of one sample (8 ounces) to a maximum of five (5) samples (40 ounces) shall be required from each lot to determine the content of binder-lubricant of the tetryl in each lot of boosters. Failure of the sample to meet the requirements specified on the applicable drawings shall be cause for rejection of the lot.

4.3.3.2 Specific gravity of booster pellet.- A sample of ten (10) booster pellets shall be selected from each lot of booster pellets to determine the specific gravity. Failure of one or more booster pellets to comply with the requirements specified on the applicable drawing shall be cause for rejection of the lot.

4.3.3.3 Process control audit.- Periodic unscheduled monitoring shall be conducted for the process operations on the following assemblies to determine compliance with the requirements of the applicable drawings.

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4.3.3.3 (Cont'd)

a. Fuze subassembly, Drawing F10542754

(1) Removal of dust cover and insertion of closing cup assembly

b. Fuze assembly, Drawing F10542753

(1) Insertion of booster pellet in closing cup.

(2) Application of sealing compound to cup threads.

(3) Tightening of booster cup on safety adapter.

4.3.3.4 Waterproofness test.- Eight fuze assemblies per F10542753 shall be selected from each lot and subjected to the waterproofness test. Evidence of water entry shall be cause for rejection of the lot.

4.3.3.5 Container airtightness.- A sample quantity of packed fuze containers shall be tested for airtightness to determine compliance with the requirements of Drawing F8864492. Sampling quantities shall be determined in accordance with MIL-STD-105, using an AQL of .40 percent or MIL-STD-1235, CPS-1 using an AQL of .40 percent.

4.3.4 Inspection equipment.- The examinations and tests shall be made using equipment listed in IEL-10542775. Equipment design shall be in accordance with the code designation on the applicable IEL.

4.4 Test methods and procedures.

4.4.1 Binder-lubricant content of tetryl.- A sample of approximately 8 ounces of tetryl shall be selected and determined as follows: An accurately weighted portion of five (5) grams of the sample shall be transferred to a tared medium porosity sintered glass filtering crucible. The sample shall be extracted with four 10-milliliter portions of acetone. Each portion shall be maintained in contact with the residue for approximately five (5) minutes before applying suction. Any lumps which may persist shall be broken up. The side of the crucible shall be washed down with each 10 milliliter portion of acetone to insure complete removal of the tetryl. The residue shall be aspirated until the odor of acetone is no longer detectable. The crucible and contents shall be placed in a drying oven at 100 degrees (°) to 105° Centigrade (C) for ½ hour, cooled in a desiccator and weighed. The above extraction, drying and weighing procedure shall be repeated with individual 10 milliliter portions of acetone until the loss in weight on subsequent extractions totals less than 0.005 gram. The weight of the residue in the crucible shall be calculated to the percentage of total binder-lubricant in the sample to determine compliance with the requirements specified on the applicable drawing.

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4.4.2 Specific gravity and booster pellet.- The representative booster pellets shall be cleaned so as to leave no sharp corners and shall then be accurately weighed on a laboratory balance. To determine the displacement, an apparatus (A8648248) consisting of a glass beaker with ground top, a piece of flat glass of sufficient size to completely cover the top of the beaker, and a semihard rubber perforated plate, pierced at each corner with sharp pointed pins so that about $\frac{1}{2}$ inch of pin protrudes from each side, shall be used. The beaker shall be filled with clean mercury, the rubber plate inserted, and the mercury leveled off by pressing down the flat glass until it is even with the top of the beaker. The sample booster pellets shall be placed beneath the prongs of the rubber plate and the mercury leveled as before. The excess mercury shall be caught in a clean bowl. The mercury displaced by the pellet shall be calculated as follows to determine compliance with the requirement specified on the applicable drawing:

$$\text{Specific Gravity} = \frac{\sigma W}{W_1}$$

Where σ = density of mercury at test temperature, in grams/cubic centimeter.

W = weight of booster pellets, in grams.

W_1 = weight of displaced mercury, in grams.

4.4.3 Waterproofness.- The test shall be performed in accordance with the procedure specified in MIL-STD-314. The fuze shall be disassembled and inspected as follows:

a. Remove booster cup, safety adapter, head, lower cap, check all visible internal surfaces for evidence of water entry.

4.4.4 Container airtightness.- The packed fuze container shall be subjected to an external air pressure of three (3) pounds per square inch, gage, for a period of not less than 15 seconds.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level "A" - Fuzes shall be preserved and packaged in accordance with Drawing F8864492.

5.1.2 Level "B" - Fuzes shall be packaged same as Level A.

5.1.3 Level "C" - Fuzes shall be packaged in accordance with Drawing C9214705.

5.2 Packing.

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5.2.1 Level "A" - Fuzes shall be packed in accordance with Drawing C8861213.

5.2.2 Level "B" - Fuzes packaged in accordance with 5.1.1, shall be palletized in accordance with Drawing C8886458.

5.2.3 Level "C" - Fuzes shall be packed in accordance with Drawing C9214704.

5.3 Marking.- Marking shall be in accordance with instructions contained in the applicable packing drawings.

5.4 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Level of preservation and packaging required (see 5.1).

c. Level of packing required (see 5.2).

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